BanasthaliVidyapith
Department of Computer Science
Course Handout: B.Tech. (IT) –VII
July – December 2026

Date: 27-June-2026

Course Code: CS 411 **Course Name: Computer Graphics**
Credit Points: 4 **Max. Marks : 100 (CA: 40+ ESA: 60)**
Course Instructor:
Dr. Aditi Paul, Assistant Professor, IT VII Semester

Course Outcomes:

After successful completion of the course students will be able to:

CO1 Apply knowledge of mathematics, science, and engineering to implement basic transformation and scan conversion.

CO2 Apply the knowledge of co-ordinate system that is necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.

CO3 Apply aesthetic judgments and critical thinking skills to art and graphics related issues.

CO4 Recognize and evaluate critical and aesthetic issues within computer graphics and the mixed media.

Syllabus:

Section A

Components of Graphics Systems: Display devices - Refresh CRTs, Random scan and Raster scan monitors, color CRT monitors, DVST, Plasma-panel displays, Hard copy devices-printers, plotters. Displayprocessors-Random scan systems, Raster scan systems. Interactive Input devices: keyboards, touch panels, light pens, tablets, joysticks, trackball, mouse. Logical classification - Locator, Stroke, String, Valuator, Choice, Pick devices. Interactive picture construction techniques – positioning methods, constraints, grids, gravity field, rubber band methods, sketching, dragging.

Section B

Output primitives: Points and lines, DDA and Bresenham's line drawing algorithm, Antialiasing lines. Bresenham's circle drawing algorithm. Character generation. Area filling: Scan line, Boundary-fill, Flood-fill algorithm. 2-D Transformations: Basic Transformations, General Transformation equations, Reflection, Shear. Windowing and clipping: Windowing concepts, Line, Area, Text clipping algorithms, Window toView port Transformation. Segmentation: Concepts, Segment files, segment attributes.

Section C

3D Transformation: 3D co-ordinates. Basic 3D transformations. Rotation about arbitrary axis. Reflection, shear, viewing transformation. Curved lines and Surfaces: Polygon surface, Bezier Curves and surfaces, spline curves and surfaces. Fractals Geometry Methods: Introduction. Hidden surface and hidden line removal: Classification of algorithms, Back-face removal, Depth buffer method, Scan line method, Depth sorting method, Area subdivision method. Comparison. Shading: Constant intensity, Gouraud shading, Phong shading, Ray-tracing algorithms.

Suggested Books:

R1. Hearn Donald & Baker M. Pauline "*Computer Graphics- C Version*" Pearson Education
R2. Foley, Van Dam, Feiner, Hughes "*Computer Graphics Principles &Practice*"Addison Wesley
R3. Harrington, S. "*Computer Graphics: A programming Approach*" McGraw Hill International Editions
R4. Rogers, David F. "*Procedural Elements for Computer Graphics*" Second Edition, McGraw Hill.
R5. Rogers, David F. & Adams J. Alan "*Mathematical Elements for Computer Graphics*" Second Edition, McGraw Hill

Suggested E-Learning Material:

E1. http://nptel.ac.in/courses/106106090

E2. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-837-computer-graphics-fall-2012

Assessment Schedule:

Component	Marks	Submission/ Examination date	Allotment
Home assignment I**	10	27August, 2026	Topics shall be allotted in the class by 5 August 2026
Periodical test I	10	9-12September, 2026*	Lecture No. 1 to 18
Home assignment II**	10	7 October, 2026	Topics shall be allotted in the class by 14September, 2026
Periodical test II	10	30October, 2026 to 2 November, 2026*	Lecture No. 19 to 40
Semester Examination	60	2-19December, 2026*	Lecture No. 01 to 50 (Entire Syllabus)

*subject to change

** Assignment marks will be based on written document(s)/ any other component(s) as decided by the instructor(s).

Lecture wise Plan:

Lect. No.	Topics to be covered	References
	Section-A	
1-3	Introduction to CG, Applications of CG, Interactive versus Non-interactive CG, Conceptual Framework of Interactive CG, Classification of Output Devices.	R1,R2
4-5	Raster Scan Systems: Horizontal Retrace, Vertical Retrace, Interlaced Raster Scan Systems, Refresh Rate, Refresh Buffer, Random Scan Systems: Display list, Refresh Display list, Raster Scan Systems Versus Random Scan Systems. Color CRT Monitors: The Beam Penetration Method, Shadow Mask Method	R1
6-8	Non-refreshing CRTs: DVST (Direct View Storage Tube), Flat Panel Displays: Emissive (Plasma Panels, Thin Film Electroluminescent, LEDs), Non-Emissive: LCD. Architecture of Raster Scan Device, Architecture of a Random Scan Device, Color Lookup Table for raster scan devices: N-bit Plane Gray Level Frame Buffer, N-bit Plane Color Frame Buffer	R1, E1
9-10	Logical classification of Input Devices-Locator, Stroke, String, Valuator, Choice, Pick devices. Interactive Picture construction techniques-Basic positioning methods, constraints, grids, gravity field, rubber band methods, sketching and drawing, dragging.	R1
11-12	Interactive input devices: keyboards, touch panels, light pens etc.Hard Copy Output Device: Printers and Plotters	R1, R4
	Section-B	
13-15	Output Primitives: Point, Line, Circle, Curve. Line Drawing Methods: Slope Intercept Method, DDA Method Bresenham's Line Drawing Method, Numerical Based on Line Drawing Algorithms	R1
16-19	Circle Drawing Methods: Polynomial Equation, Polar Parametric Equation, Bresenham's Method, Midpoint Method, 8-Way Symmetry of the Circle. Aliasing Effects: Stair Step, Unequal Brightness, The Picket Fence Problem: Anti-aliasing Methods	R1
20-21	Area Filling: Scan Line Approach, Seed Fill Approach. Scan-Line Filling Algorithm, Boundary Fill Algorithm, Flood Fill Algorithm	R1
22-25	2D-Transformations: Introduction, Translation, Rotation, Scaling, Reflection, Shearing. Rotation - About Origin, About Pivot Point, Scaling - About Origin, About Pivot Point, Reflection – about an arbitrary line, Shearing – about origin, about pivot point. Simple 3D Transformations: Translations, Rotations (About x, y and z axis), Scaling about origin and fixed point, Shearing along x, y, and z-axis), Reflection.	R1, R5, E1

26-27	Window to Viewport Transformation: World Coordinate System, Device Coordinate System, Normalized Device Coordinate System. What is window? What is Viewport? Window to Viewport Transformation, Applications of Window to Viewport Transformation: Panning, Zooming.	R1, R2
28-31	Clipping: Concept of Clipping, Point Clipping, Line Clipping: Cohen Sutherland Line Clipping, Cyrus Beck Line Clipping, Liang Barsky Line Clipping. Polygon clipping—Sutherland-Hodgeman, Text Clipping	R1, R5
32	Segmentation: Concepts, Segment files, attributes	R3
Section-C		
33-35	3D coordinates, Basic 3D transformations, Rotation about arbitrary axis, Reflection, shear, viewing transformation	R1, R5
36-39	Curved lines & surfaces-Polygon surface, Bezier & Spline curves and surfaces	R1, R5, E1
40-43	Hidden surface and Hidden line removal algorithms: Back-face, Depth-buffer, Scan-line, Depth-sorting, Area-subdivision, and comparison among these algorithms	R1, R2
44-47	Projection: Perspective & Parallel	R1, R5
48-49	Shading: Constant intensity, Gouraud shading, Phong shading, Ray tracing algorithms	R1
50	Fractals Geometry methods: Introduction	R1

Dr. Aditi Paul